MANU GUPTA

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Summary

An undergraduate student with strong **problem-solving skills** and knowledge of **DSA, OOPs, DBMS**. Passionate about building **secure, scalable backend systems** and full-stack applications to contribute to quality products.

Skills

Programming Languages: Java, Python, C++, C, SQL, JavaScript, TypeScript
Frameworks & Libraries: Spring Boot, FastAPI, React.js, Streamlit, Google OR-Tools, Pytest
Databases & Tools: PostgreSQL, SQLite, Docker, Git, GitHub, k6, Flyway, Testcontainers, Railway, Vercel
Coursework: Data Structures and Algorithms, OOPs, DBMS, Computer Organization and Architecture

Education

B.Tech in **Computer Science and Engineering** — CGPA: 7.7 | 2024 – 2028
Galgotias University, Greater Noida

Senior Secondary | CBSE (Class XII) — Percentage: 71 | 2023 – 2024
Vivekanand School, Narela

High School | CBSE (Class X) — Percentage: 91 | 2021 – 2022
Delhi Mar Thoma Public School, Ghaziabad

Projects

Medicity – Clinic and Pharmacy Platform *Sept 2025 – Oct 2026*
Tech Stack: Java 21, Spring Boot 3, React 18, TypeScript, PostgreSQL, Docker

- Secured the REST API with **short-lived JWTs, single-use refresh tokens**, login throttling, row-level authorization and a trigger-enforced, append-only audit log.
- Eliminated double booking using a **PostgreSQL partial unique index**; a k6 load test with 200 concurrent attempts on 20 slots produced **exactly 20 bookings**.
- Built a full-stack platform where patients book doctors, store prescriptions and reserve medicines at verified chemists, with role-based workflows for patients, doctors, chemists and admins.
- Deployed the API on **Railway** and the frontend on **Vercel** using Docker, with Flyway migrations and Testcontainers integration tests.
- Integrated **Claude and Gemini APIs** to read handwritten prescription photos, and added WebSocket video visits and SMS/email login codes.

LedgerLens – AI Reconciliation Controller *Aug 2026 – Sept 2026*
Tech Stack: Python, FastAPI, React, Groq LLM, Pytest

- Built a three-stage pipeline that matches payment settlements to bank statements, handling **MDR + GST deductions, T+2 settlement lag**, truncated references and batched nodal-account credits.
- Added an **LLM agent** for ambiguous matches and a live React dashboard showing each decision's confidence, match rate, exceptions and throughput.

RailBlock – Automatic Block Planning for Indian Railways (SIH 2026) *Sept 2026*
Tech Stack: Python, Google OR-Tools (CP-SAT), FastAPI, React, TypeScript, Streamlit, SQLite

- Built a scheduler that models three maintenance departments' track-block requests as a constraint problem and uses **OR-Tools CP-SAT** to return a **provably optimal, conflict-free plan** (6 sections, 18 requests in ~120 ms).
- When requests cannot all fit, isolates the **minimal conflicting set** via a deletion filter and explains it; an independent **verify()** re-checks every plan against all rules.
- Developed role-based Streamlit and React/FastAPI interfaces with SQLite persistence, CSV export and a **17-test** suite.

Certifications

Smart India Hackathon 2025 – Pre-Qualifier (Certificate of Participation) *Sept 2025*

Achievements

- **Qualified** for **Smart India Hackathon (SIH) 2025** – ranked among the top **130** teams out of **500+** in the Pre-Qualifier.
- Solved **150+** DSA problems on **LeetCode** and **GeeksforGeeks** covering arrays, trees, graphs and dynamic programming.